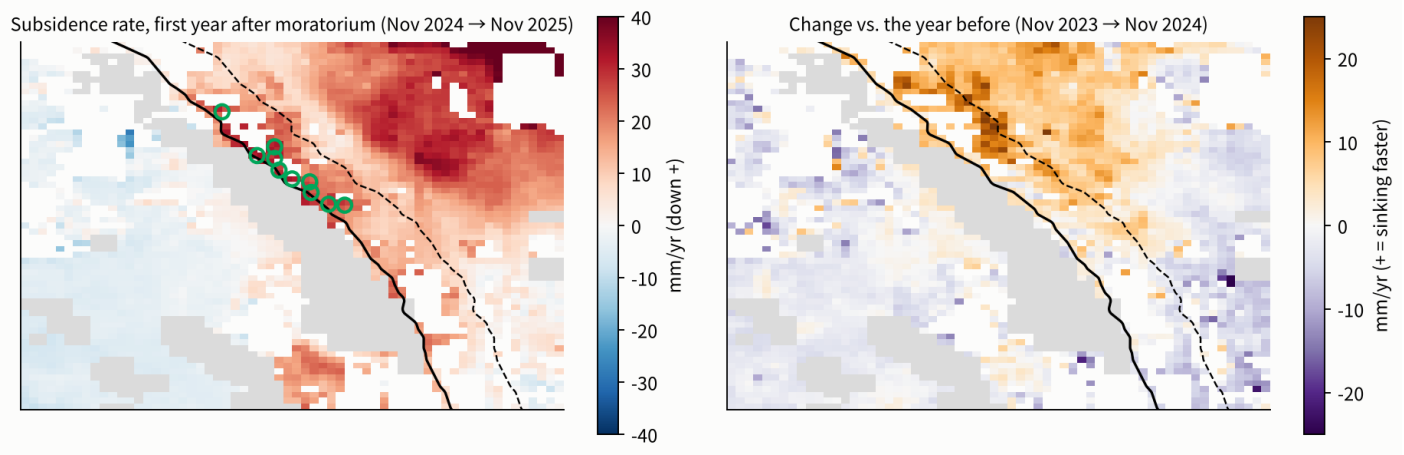
AquaLimit: satellite-informed groundwater pumping decisions

California Aqueduct, Mileposts 195–215 (Lost Hills). Student research using public data only: NASA OPERA InSAR, Landsat, gridMET, DWR well completion reports and water levels, Aqueduct GPS (Nevada Geodetic Lab), USGS CVHM2 layers, MODFLOW 6 + CSUB ensemble.

What we found	Number	Basis
With the 2.5-mile buffer moratorium, share of aqueduct points expected to stay under 0.5 in/yr in the 2026 irrigation season	**76% wet / 45% average / 32% hot-dry**	Model simulation (ensemble re-run in MODFLOW)
Cost to keep every aqueduct point under 0.5 in/yr in every ensemble member: well-by-well reductions vs. a fixed 6.9 km buffer (the smallest that holds in a hot-dry year)	**20% lower** in an average year (29% wet, 10% hot-dry)	Model simulation (both plans re-run in the MODFLOW ensemble)
First year after the moratorium vs. the year before (NASA OPERA): change in subsidence rate inside the buffer minus outside	**+2.9 mm/yr** (90%: +0.6 to +4.5): no visible improvement inside the buffer	Satellite observation only (difference-in-differences)
Data quality, 2023–2026: direction of vertical motion matching the five Aqueduct GPS stations	**OPERA 5/5, DWR TRE Altamira 3/5**; median error 8.3 vs 17.0 mm. At P544 in 2020–22, TRE already shows half the GPS subsidence (13.6 vs 27.8 mm)	Observation (same 500 m cells)
Aqueduct GPS shows a seasonal cycle (up in summer, down in winter), so rates should be compared over whole years	Q204: 15.4 to 15.9, Q208: 20.9 to 21.9 mm/yr (year before vs. year after)	Observation (Nov-to-Nov)



NASA OPERA vertical velocity. Solid line: California Aqueduct; dashed: 4 km (2.5 mi) buffer boundary on the farm side; grey: oil fields (excluded); green circles: 1 km cells inside the buffer still sinking fastest after December 2024 (25–33 mm/yr), suggested as places to check flowmeters first. These cells also include delayed clay compaction, so they are screening candidates, not evidence of violations.

What AquaLimit would provide a GSA: (1) whether the current rule is enough to meet a subsidence threshold under each weather and allocation scenario, (2) the lowest-cost well-by-well reductions that meet it, (3) a monthly satellite screening list for compliance checks, and (4) a data-quality check of satellite products against GPS.

Limitations: well-level metered pumping is not public, so pumping is estimated from satellite evapotranspiration and calibrated; the 0.5 in/yr threshold is Wheeler Ridge-Maricopa's; the model still forecasts 3-year cumulative subsidence less accurately than simple trend extension (21.2 vs 7.2 mm). All policy numbers are simulations.